

Matrix Theory

1. Let

$$M = \begin{bmatrix} b_1 + c_1 & b_2 + c_2 & b_3 + c_3 \\ c_1 + a_1 & c_2 + a_2 & c_3 + a_3 \\ a_1 + b_1 & a_2 + b_2 & a_3 + b_3 \end{bmatrix} \quad \text{and} \quad M_0 = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}.$$

Show that M is row equivalent to M_0 by finding a sequence of elementary row operations that when applied to M result in M_0 . Also compute the product E of the corresponding elementary matrices so $M_0 = EM$.

2. (variation on NS 3.2.2c) Let

$$A = \begin{bmatrix} 0 & 2 & 3 & 7 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- (a) Apply the algorithm from Section 3.2 to reduce A to row echelon form.
- (b) Find $\text{rank}(A)$.

